

Sikaflow – Windows 11 VM Network Assessment

Cybersecurity Internship – Week Focus: Reconnaissance & Network Enumeration

Cybersecurity Group: 3

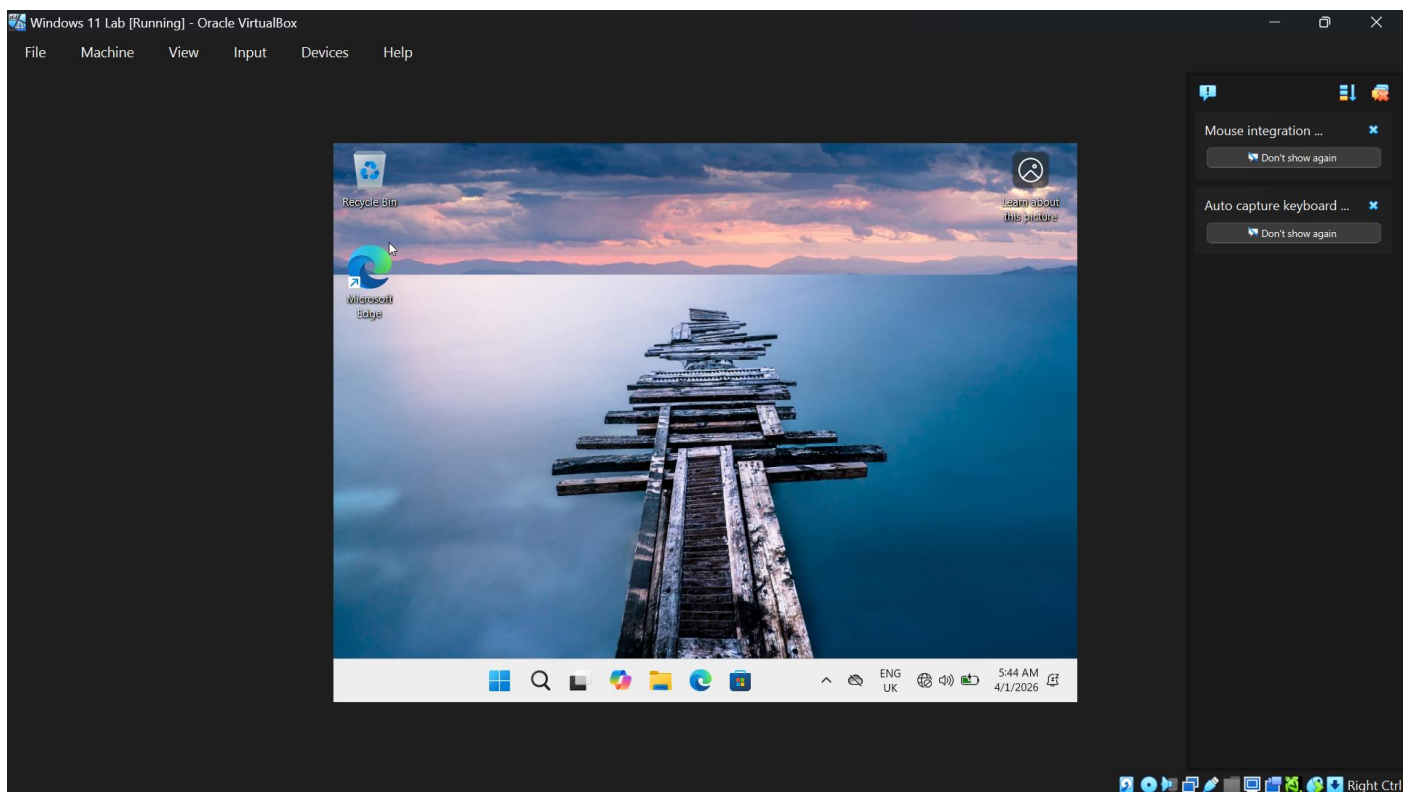
Date: 1st April, 2026

1. Objective

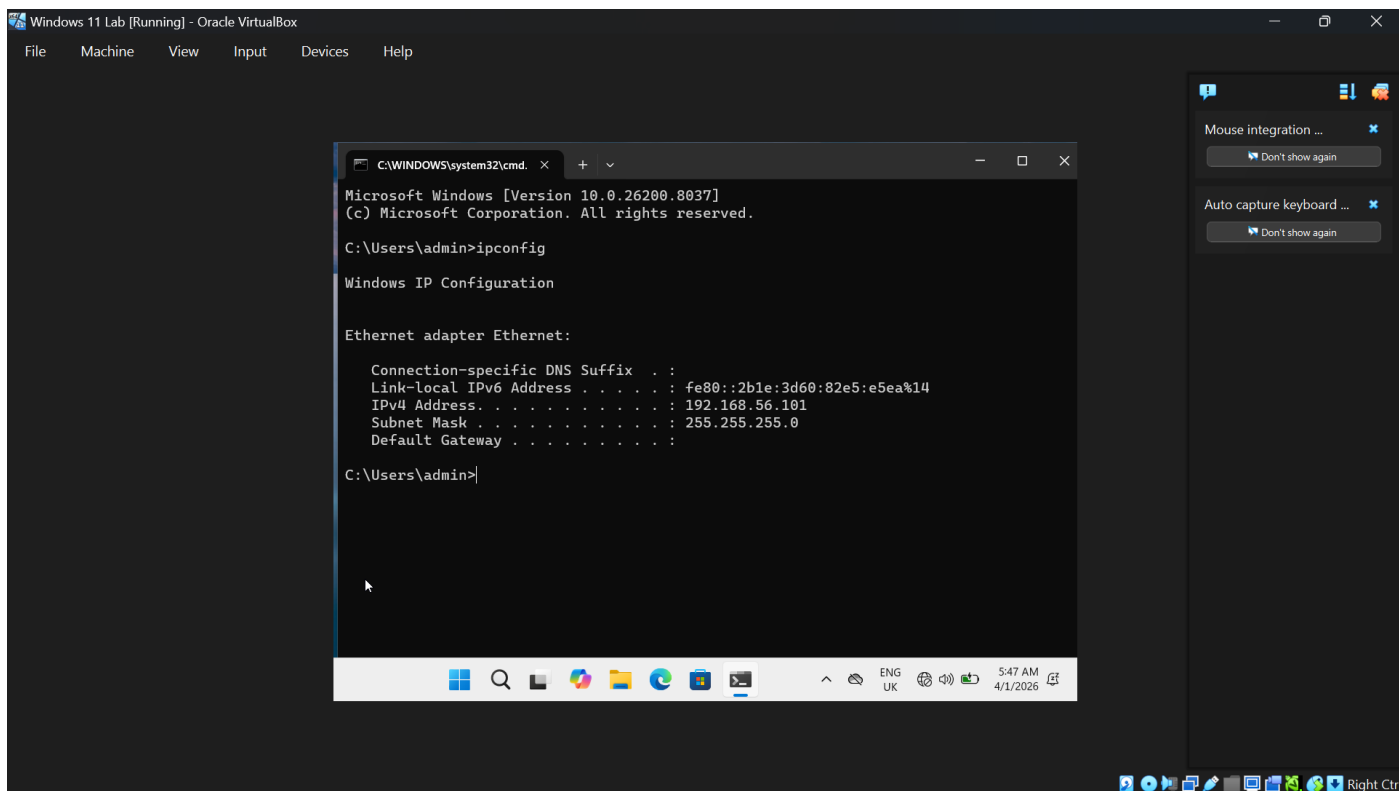
The goal of this assessment was to perform network reconnaissance and service enumeration on a Windows 11 system, simulating a target environment for a fintech organization, **Sikaflow**, based in Ghana. The purpose was to identify live hosts, open ports, exposed services, and potential vulnerabilities, and to evaluate their impact on business operations.

2. Host Discovery

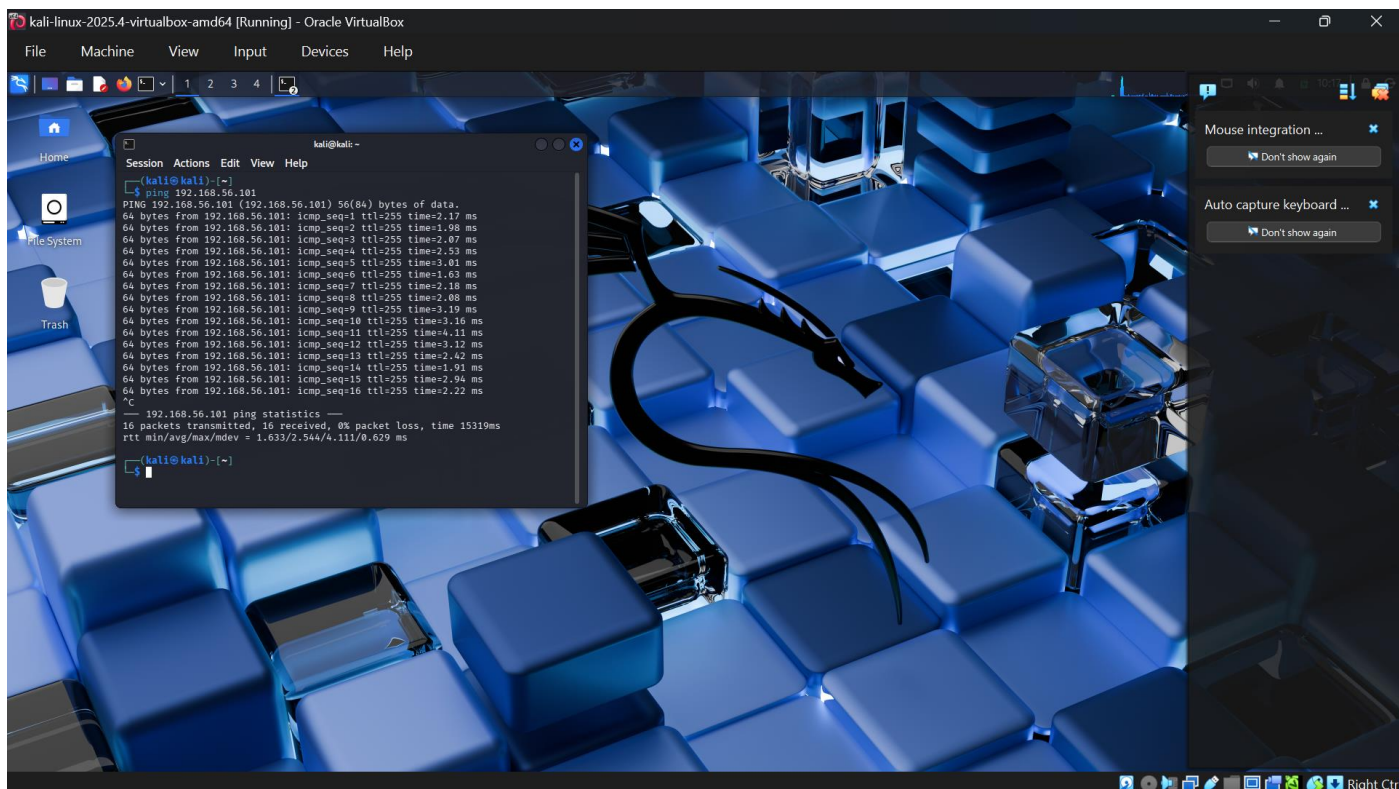
- **Target IP:** 192.168.56.101
- **Command:** ping 192.168.56.101



A screenshot of Windows 11 successfully installed and running on Oracle Virtual Box



A screenshot of the IPv4 address of Windows 11 using the ipconfig command (192.168.56.101)



A screenshot showing a successful network communication between Kali Linux and Windows.

Interpretation:

- Windows VM is **alive** and reachable from Kali Linux
- Network configuration between VMs is correctly set up

3. Port Scanning Results

Service Enumeration

Command: nmap -sV -sC 192.168.56.101

Open Ports Detected:

Ports	Service	Description
135	MSRPC	Standard Windows RPC Service
139	NetBIOS-SSN	Hostname exposed: DESKTOP-JTBP7L5
445	Microsoft-DS	SMB2, message signing enabled

Interpretation:

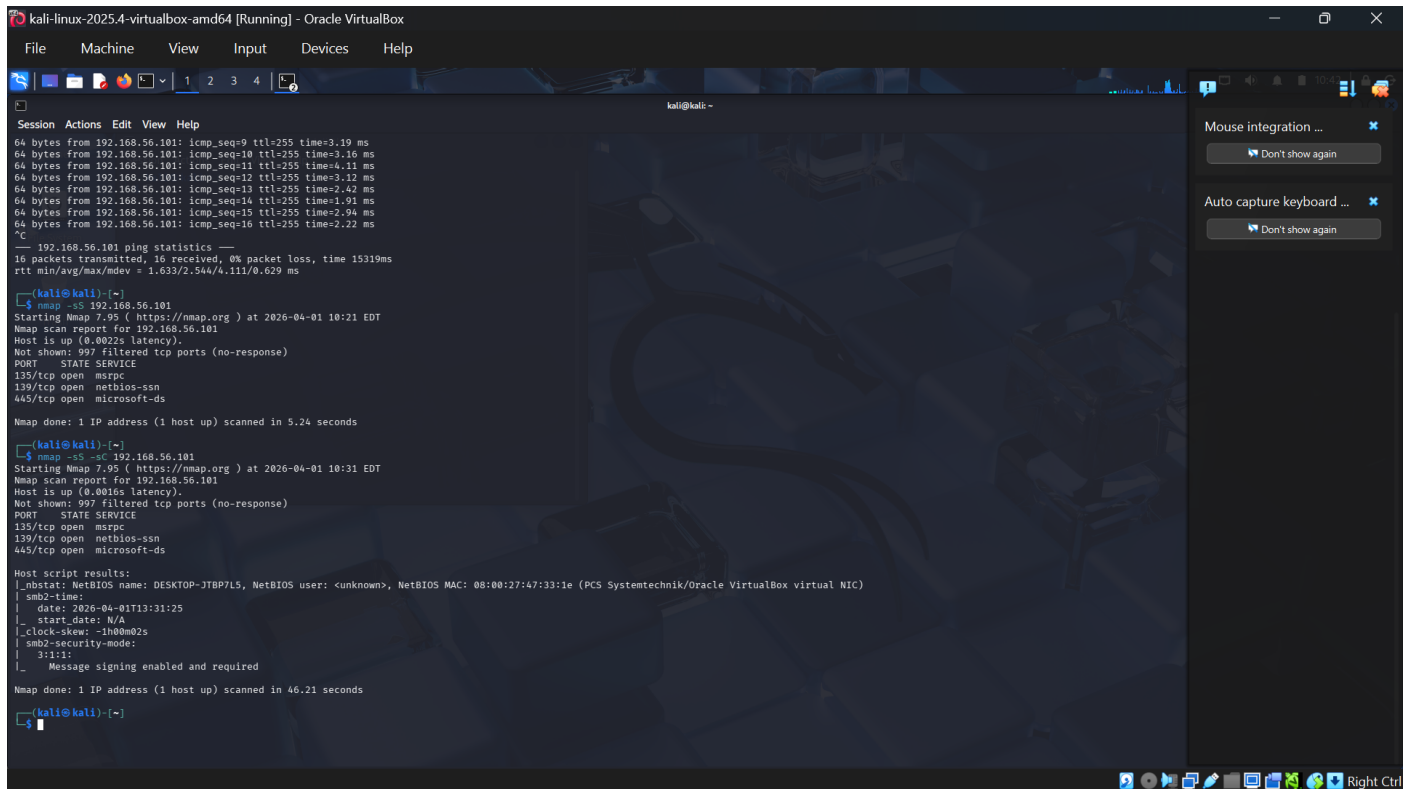
- Exposed services contribute to the **attack surface**
- SMB (port 445) is particularly sensitive for file sharing and potential exploitation

Host Script Results:

- NetBIOS Name: DESKTOP-JTBP7L5
- NetBIOS User: <unknown>
- MAC Address: 08:00:27:47:33:1e (Oracle VirtualBox)
- SMB2 Security Mode: Message signing enabled
- SMB2 Time: Clock skew -1 hour

Interpretation:

- Hostname and MAC address are visible → **information leakage risk**
- SMB signing enabled → **positive security measure**
- Clock skew could affect logging/authentication accuracy



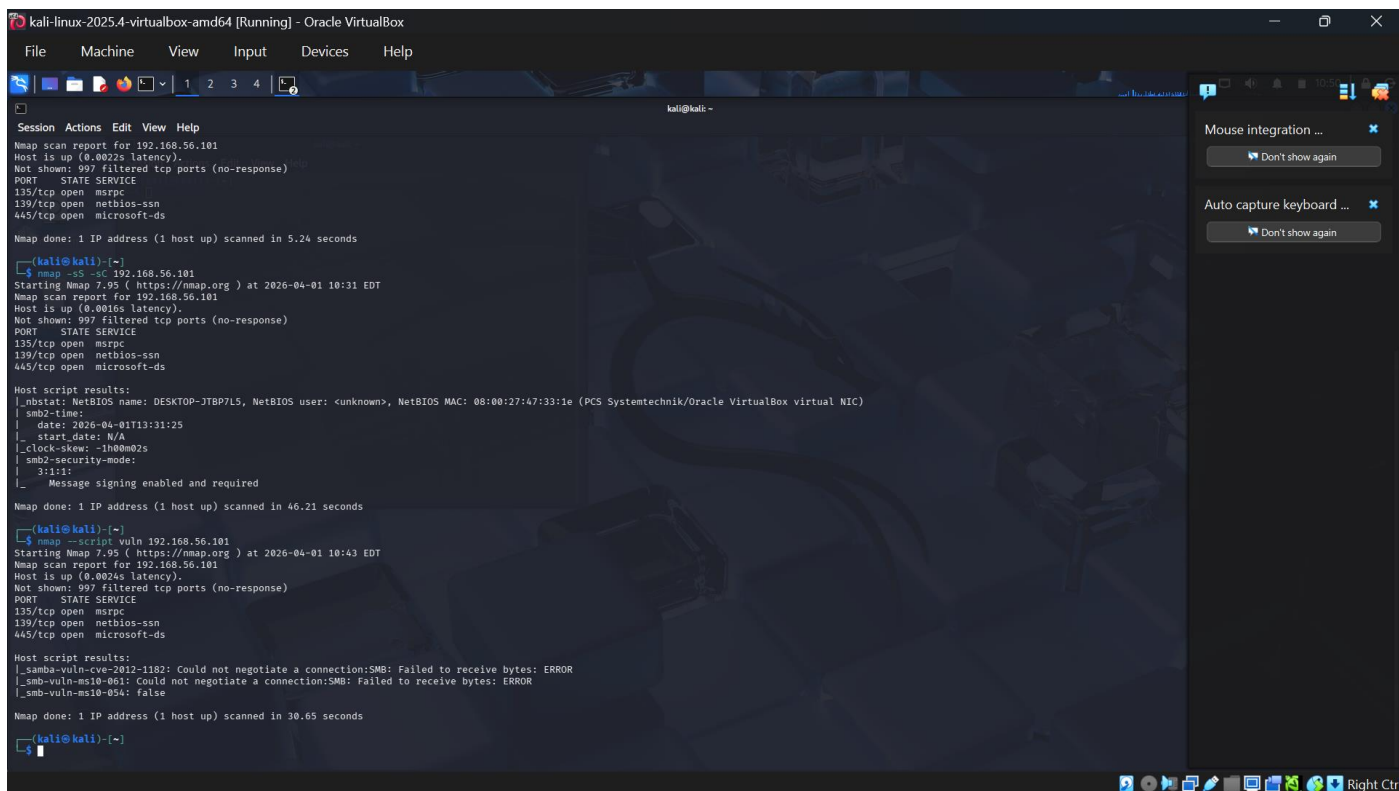
A screenshot showing the nmap scan revealing several open ports and host script results.

4. Vulnerability Assessment

Command: `nmap --script vuln 192.168.56.101`

Findings:

- No known SMB vulnerabilities detected (smb-vuln-ms10-054: false)
- Some scripts failed to receive responses (likely firewall/security settings)



A screenshot showing vulnerability scan results

Interpretation:

- Windows 11 is **patched and hardened**
- Exposed services still present a **potential attack surface** for enumeration or brute-force attacks

5. Business Impact Analysis - SikaFlow

Context: Sikaflow relies on secure, available systems for financial transactions and customer data.

Port/Service	Potential Risk	Impact on Business
445 SMB	Unauthorized file access	Exposure of financial data, internal documents
139 NetBIOS	Enumeration attacks	Attackers can map internal systems
135 MSRPC	Remote exploitation	Lateral movement attacks possible
3389 RDP (if exposed)	Unauthorized remote access	System takeover, fraudulent transactions

Insight:

Even without direct vulnerabilities, exposed services increase the **attack surface** and can affect **confidentiality, integrity, and availability**.

6. Conclusion

The Windows 11 VM shows multiple open services that can be enumerated over the network. SMB (port 445) and NetBIOS (port 139) are particularly critical. No known vulnerabilities were detected, but information leakage (hostname, MAC address, SMB shares) still presents a **risk to the organization**.

For a fintech company like **Sikaflow**, this assessment emphasizes the importance of:

- Minimizing exposed services
- Implementing strong access controls
- Continuous monitoring of network activity

Reconnaissance and Network Enumeration Report

Company: SikaFlow (Fintech Startup – West Africa)

Cybersecurity Group: 3

Week: 2

Assessment Type: Internal Lab Security Assessment

Date: 2nd April, 2026

1. Executive Summary

This report documents the reconnaissance and network enumeration activities performed within a controlled lab environment as part of the cybersecurity internship program at **SikaFlow**.

The objective of this exercise was to simulate the early stages of a penetration test by identifying active hosts, discovering open ports, enumerating running services, and assessing potential security risks associated with exposed services.

The assessment was conducted using **Kali Linux** against the intentionally vulnerable system **Metasploitable 2**, running within **Oracle VM VirtualBox**.

The scan identified **multiple exposed services**, including FTP, SSH, Telnet, HTTP, MySQL, and others. Several of these services are outdated or insecure and could potentially allow unauthorized access or compromise of sensitive financial infrastructure if deployed in a real fintech environment.

2. Objective

The primary objectives of this exercise were to:

- Establish a virtual cybersecurity lab environment.
- Discover live hosts on the network.
- Identify open ports and running services.
- Enumerate service versions and technologies.
- Identify potentially vulnerable or exposed services.
- Analyze the potential business impact on fintech infrastructure.

3. Lab Environment Setup

The testing environment consisted of virtual machines configured to simulate a real-world penetration testing scenario.

Component	Description
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Virtualization Platform	Oracle VM VirtualBox
Network Configuration	Host-Only Adapter

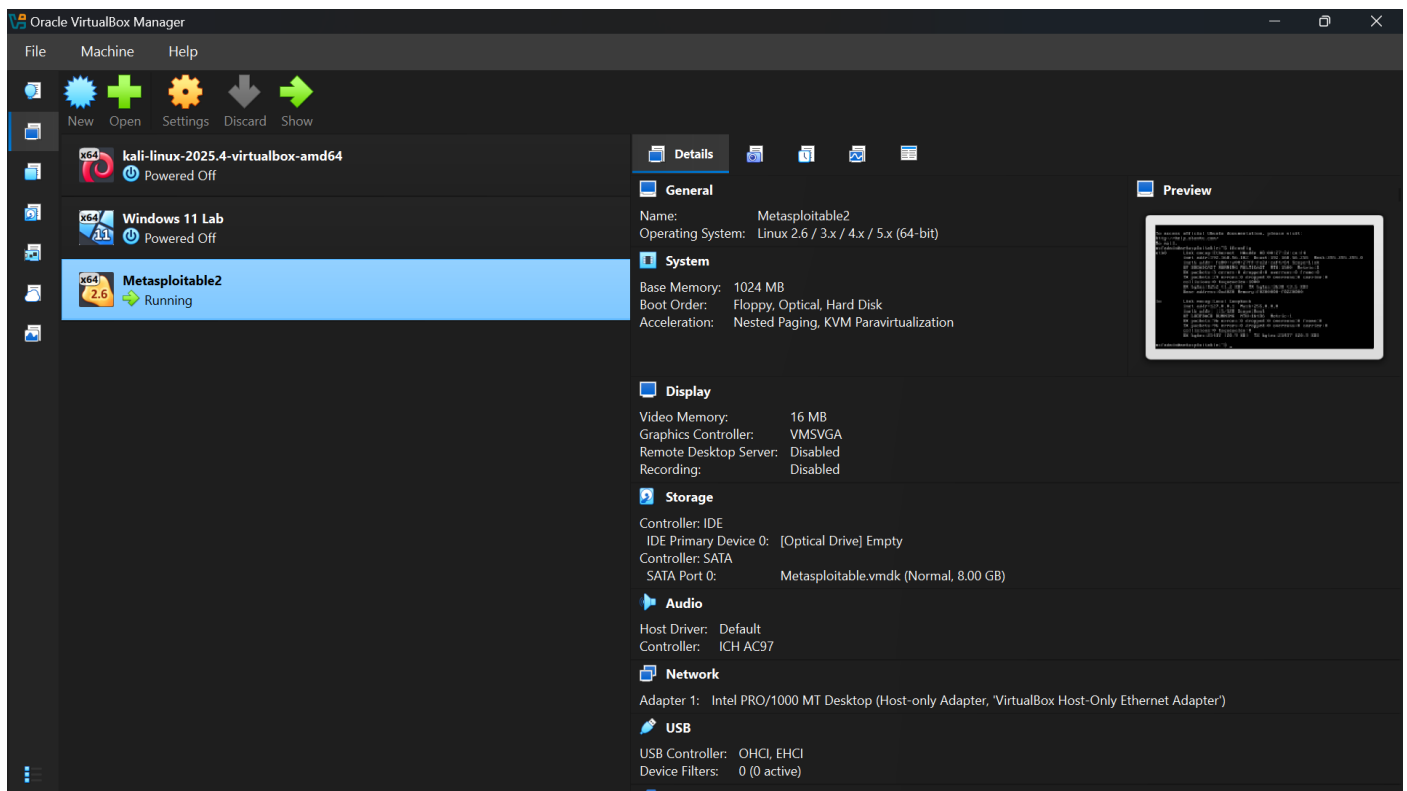


Figure 1. Virtual Lab Environment Setup

4. Target Identification

The IP address of the Metasploitable system was identified using the `ifconfig` command.

Command used: `ifconfig`

Identified IP address: 192.168.56.102

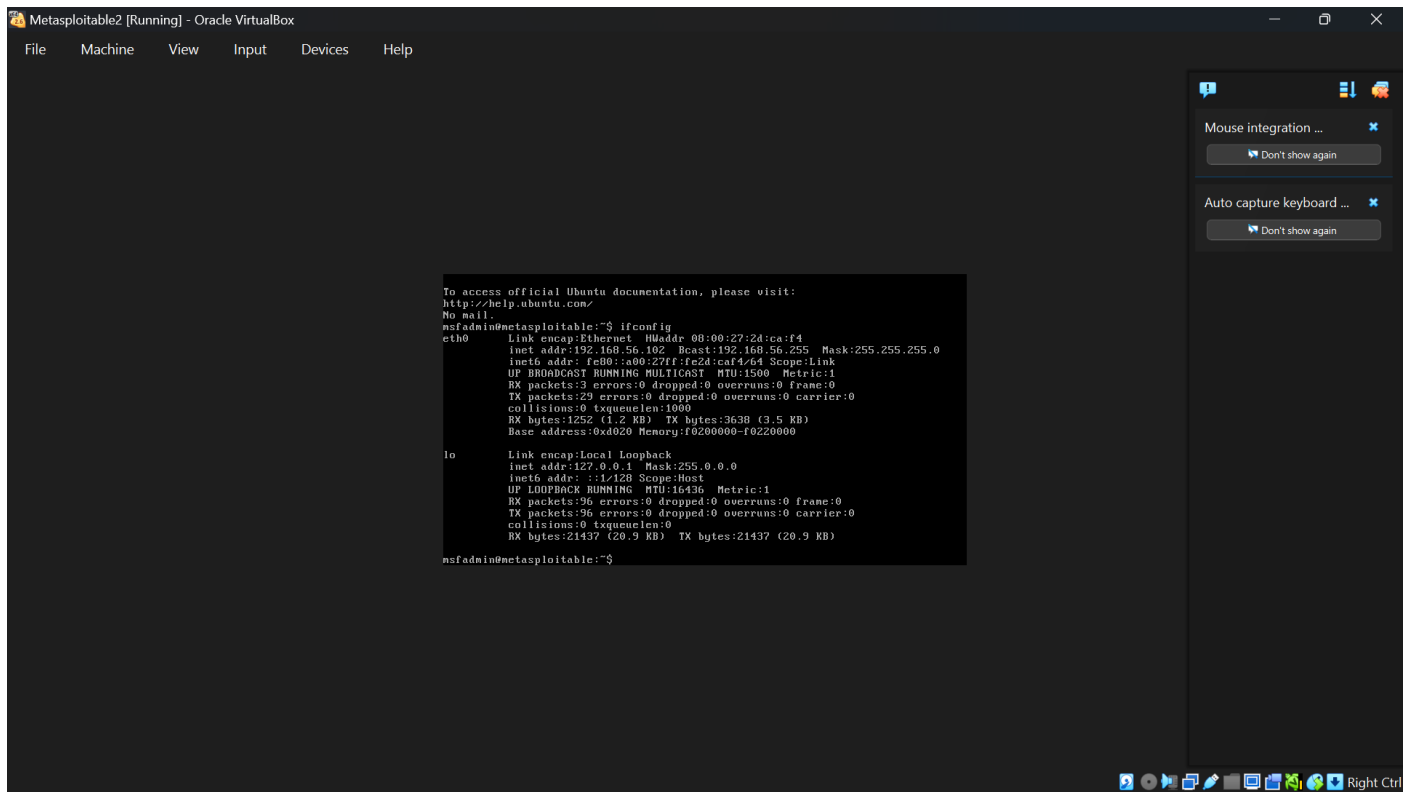


Figure 2: Target Machine IP Address Identification (inet addr: 192.168.56.102)

5. Host Discovery

A ping scan was performed to identify active hosts within the network.

Command used in Kali Linux: ping 192.168.56.102

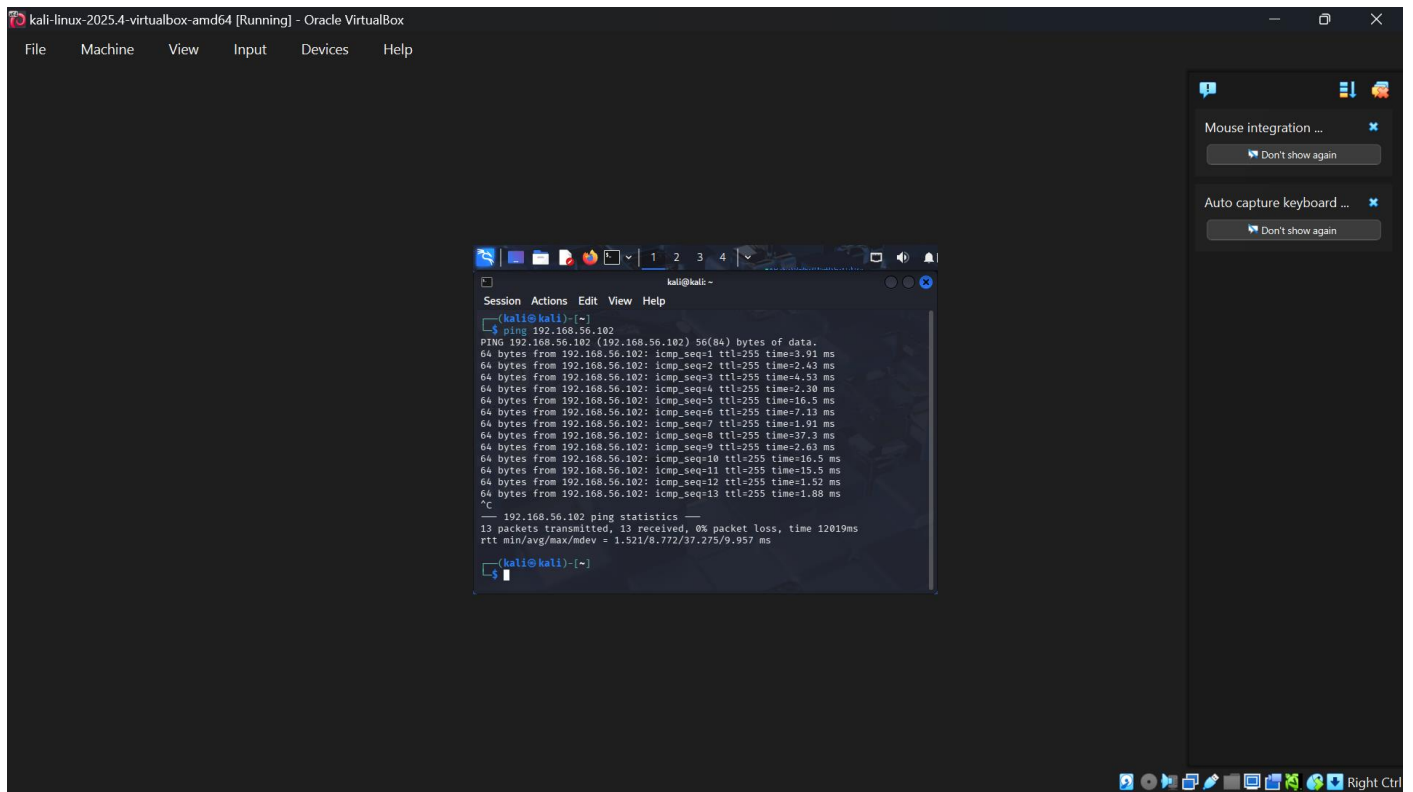


Figure 3: A screenshot showing a successful communication between Kali Linux and Metasploitable 2

Command used with nmap: `nmap -sn 192.168.56.102`

Results:

Host IP	Status
192.168.56.102	Host is up

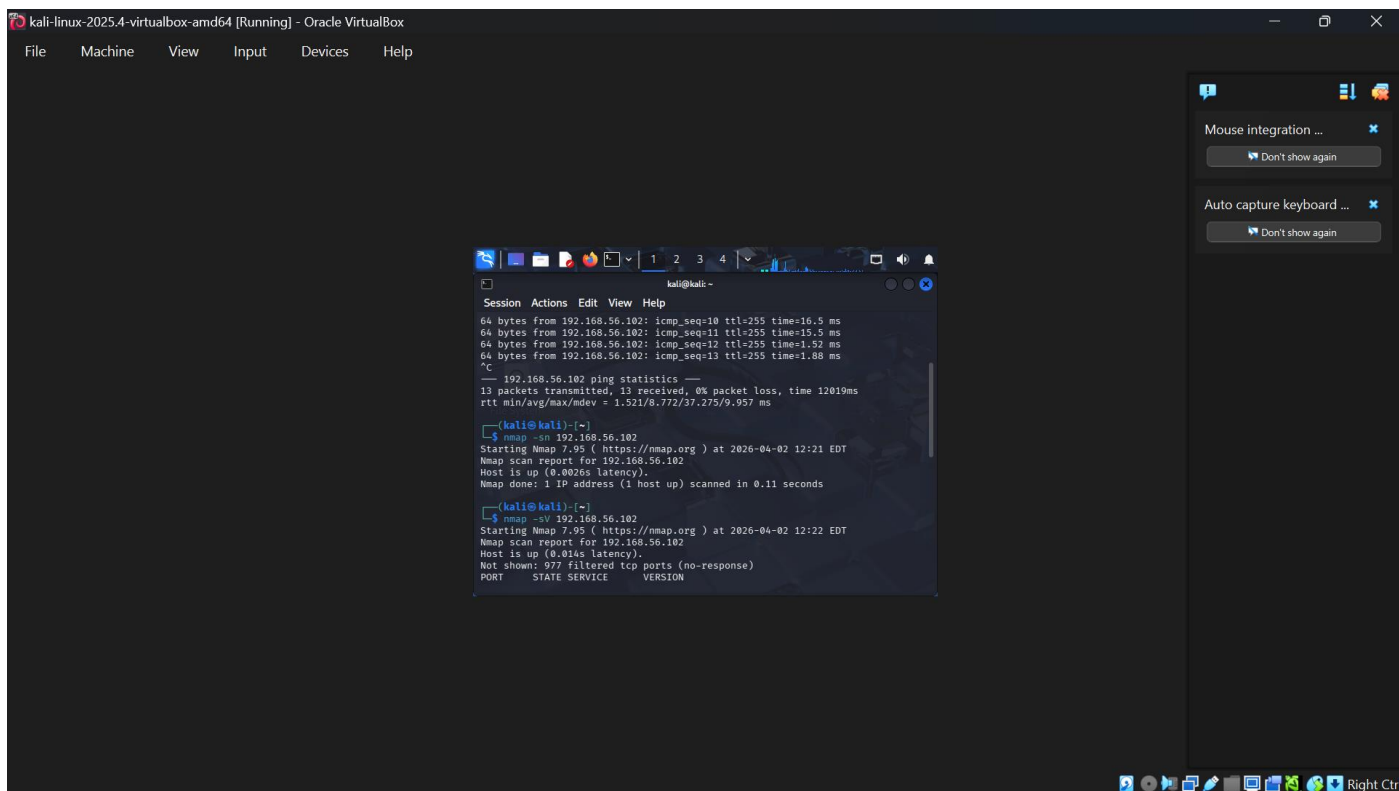


Figure 4: An nmap scan showing 1 IP address (1 host up)

6. Service Enumeration

Service enumeration was performed to identify the software versions running on the discovered ports.

Command: `nmap -sV 192.168.56.102`

Identified Services

PORT	SERVICE	VERSION
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7
23	Telnet	Linux Telnet
25	SMTP	Postfix smtpd

80	HTTP	Apache 2.2.8
3306	MySQL	MySQL 5.0
5432	PostgreSQL	PostgreSQL 8.3
5900	VNC	VNC protocol

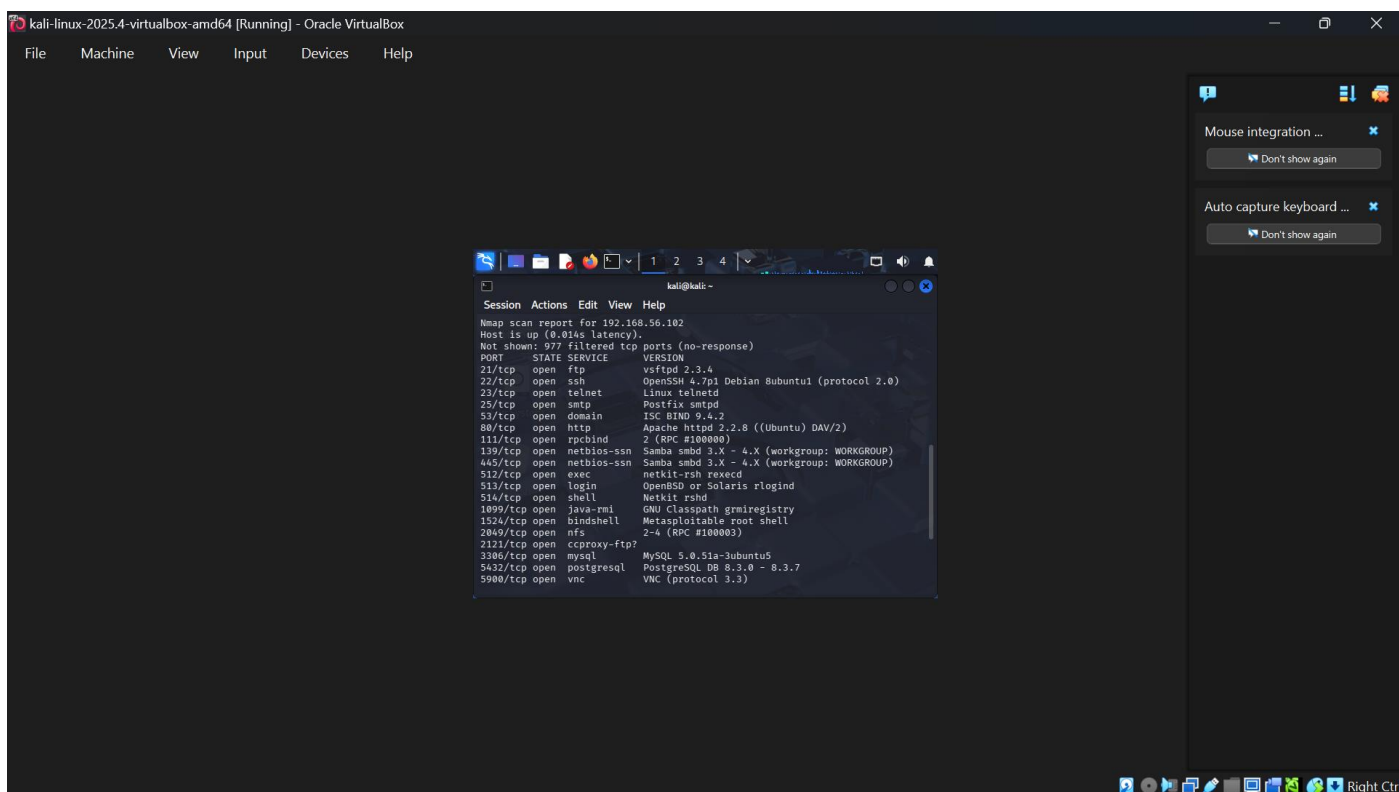


Figure 5: Service Version Enumeration 1

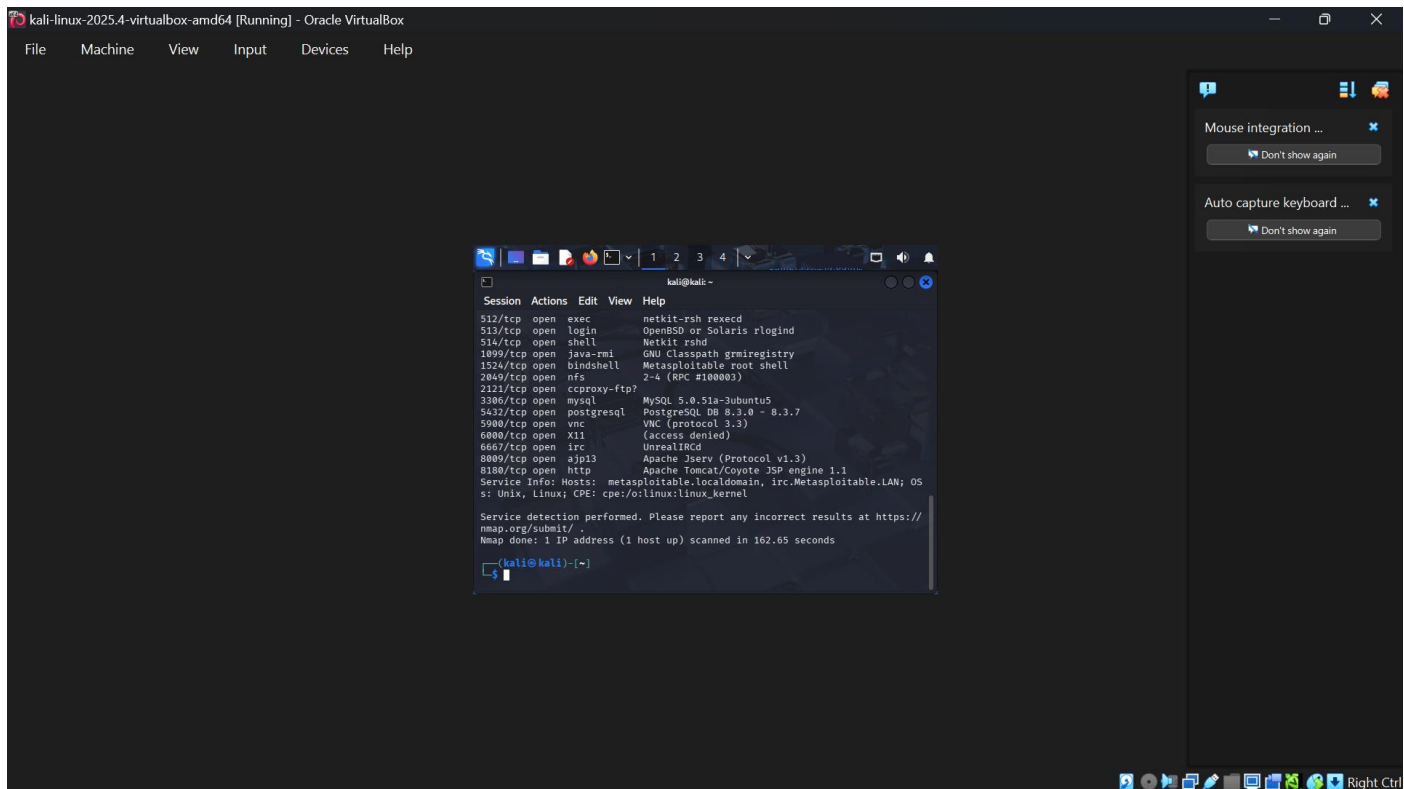


Figure 6: Service Version Enumeration 2

7. Identified Security Risks

Several services discovered during enumeration present potential security risks.

Service	Risk Level	Description
FTP (vsftpd 2.3.4)	Critical	Known backdoor vulnerability
Telnet	High	Credentials transmitted in plaintext
Apache 2.2.8	High	Outdated web server vulnerable to exploits
SMB	High	Can be exploited for lateral movement
MySQL	Critical	Potential exposure of financial databases
PostgreSQL	High	Unauthorized database access risk
VNC	Medium	Remote desktop access risk

8. Business Impact Analysis (SIKAFLOW)

If similar vulnerabilities existed within SikaFlow's production infrastructure, the potential consequences could include:

Financial Risks

- Unauthorized transactions
- Manipulation of financial data
- Theft of digital assets

Operational Risks

- Disruption of payment services
- System downtime
- Ransomware attacks

Regulatory Risks

Fintech platforms must comply with strict data protection and financial regulations. Security breaches could lead to regulatory penalties and compliance violations.

Reputational Risks

Loss of customer trust could significantly affect the company's growth and investor confidence.

9. Security Recommendations

To mitigate these risks, the following security practices are recommended:

1. Disable unnecessary services such as Telnet and IRC.
2. Replace Telnet with secure alternatives such as SSH.
3. Regularly update and patch software services.
4. Restrict network access using firewall policies.
5. Implement network segmentation.
6. Use intrusion detection and monitoring systems.
7. Perform regular vulnerability assessments and penetration testing.

10. Conclusion

The reconnaissance and enumeration phase successfully identified multiple exposed services within the lab environment. The exercise demonstrates how attackers can gather critical information about network infrastructure during the early stages of an attack.

These findings highlight the importance of proactive security measures, regular system updates, and continuous monitoring to protect sensitive financial systems within fintech organizations such as SikaFlow.

